

Experiment - 01

EE2501

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Task 1

Aim

To implement a high-pass RC filter with a specified cut-off frequency determined by the month of birth of lab partner_1. The objectives include:

- Implementing a single-stage and double-stage RC high-pass filter.
- Measuring and plotting the frequency response (gain and phase) over the range 10 Hz to $100f_c$.
- Determining the experimentally obtained cut-off frequency and roll-off slope.
- Comparing experimental results with SPICE simulations and determining R, C values for best fit.

Equipment and Components used

- Multimeter to check resistance and capacitance values
- Resistor of 1 M Ω
- Capacitor of 15 pF
- Digital Storage Oscilloscope (Tektronix 1072B) from Tinkerer's Lab
- Function Generator (Tektronix AFG1022)

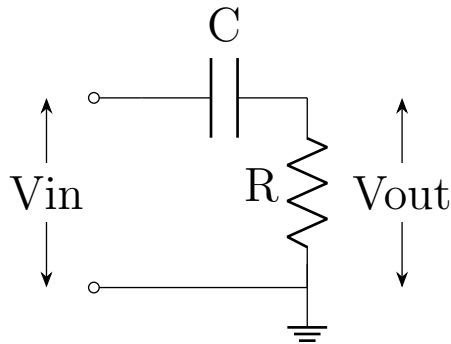
Procedure

1. Design a single-stage high-pass RC filter with cut-off frequency $f_c = \frac{1}{2\pi RC}$, where f_c (in kHz) equals the month of birth of lab partner_1(10, in our case).
2. Build the circuit on the breadboard and apply a sinusoidal input signal.
3. Vary the input frequency from 10 Hz to $100f_c$ in suitable steps.
4. At each frequency, measure input and output amplitudes and phases using the DSO.
5. Compute the gain (in dB) and phase shift for each frequency point. Determine the experimentally obtained cut-off frequency and roll-off slope.
6. Repeat the above steps for a double-stage RC high-pass filter using the same R, C values.
7. Simulate the single-stage and double-stage filters in ngspice. Compare simulated frequency responses with experimental results.
8. Tweak simulation R, C values or parameter models for best fit with experimental data.(We minimized **mean square error** to obtain the best fit)

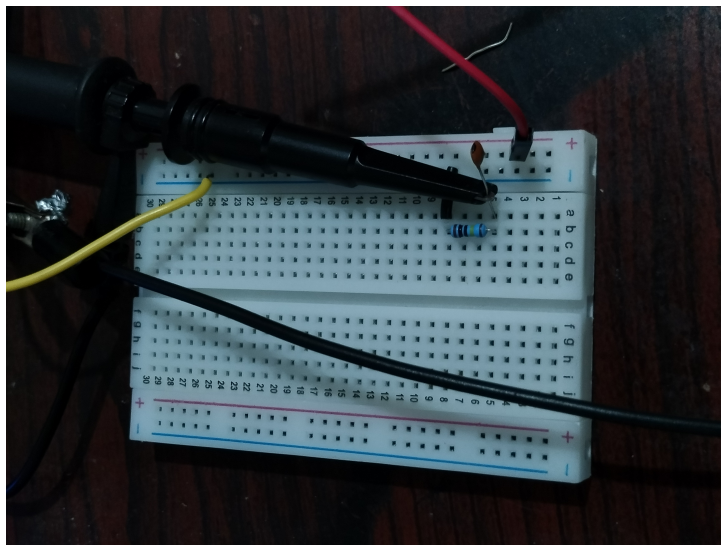
Stage 1 : Circuit Design and Setup

The circuit implemented is an RC high-pass filter with $R = 1\text{ M}\Omega$ and $C = 15\text{ pF}$. The theoretical cut-off frequency is given by

$$f_c = \frac{1}{2\pi RC} \approx 10.6\text{ kHz.}$$



Breadboard Implementation



Breadboard implementation of the RC high-pass filter.

Observations

Freq	Vout	Vin	H	H(theory)	%Error in H	20log—H—	20log—H'theo—	delT	delPhase	delPhase(Theo)	Phase error(%)
100	0	9.92	0	0.0099995	100	#NUM!	-40.00043427	2.8ms	1.759291886	1.56079666	12.71755835
300	0.208	9.84	0.021138211	0.029986509	29.50759521	-33.49863527	-30.4614818	720us	1.357168026	1.540805322	11.91826722
500	0.68	10	0.068	0.049937617	36.16989388	-23.34982175	-26.03144373	520us	1.63362818	1.520837931	7.416322705
750	0.88	10	0.088	0.074789948	17.66287057	-21.11034656	-22.52313535	280us	1.319468915	1.495936479	11.79646108
1000	1.12	10.8	0.103703704	0.099503719	4.220932367	-19.68411466	-20.04321374	220us	1.382300768	1.471127674	6.038014802
2500	2.08	9.6	0.216666667	0.242535625	10.66604478	-13.28415796	-12.30448921	70us	1.099557429	1.325817664	17.06571281
5000	5.2	10.8	0.481481481	0.447213595	7.66253225	-6.348408237	-6.989700043	28us	0.879645943	1.107148718	20.54852895
7500	4.4	9.2	0.47826087	0.6	20.28985507	-6.406703017	-4.436974992	16us	0.753982237	0.927295218	18.69016229
10000	6.4	10.8	0.592592593	0.707106781	16.19475186	-4.54487563	-3.010299957	11us	0.691150384	0.785398163	12
12500	5.28	9.2	0.573913043	0.780868809	26.50326962	-4.823078096	-2.14843848	6us	0.471238898	0.674740942	30.16002609
15000	6.8	10.8	0.62962963	0.832050294	24.32793619	-4.018296856	-1.597008429	4us	0.376991118	0.588002604	35.8861481
17500	6.2	9.2	0.673913043	0.868243142	22.38199062	-3.427922757	-1.227172766	4us	0.439822972	0.519146114	15.27954088
20000	7.2	9.2	0.782608696	0.894427191	12.50168784	-2.129106618	-0.96910013	2.4us	0.301592895	0.463647609	34.95212983
30000	7.2	9.2	0.782608696	0.948683298	17.50580017	-2.129106618	-0.457574906	2.4us	0.452389342	0.321750554	40.60250587
35000	7.6	9.2	0.826086957	0.961523948	14.08565969	-1.659484701	-0.340797896	800ns	0.175929189	0.278299659	36.78426009
50000	6.6	9.4	0.70212766	0.980580676	28.39674726	-3.071678361	-0.170333393	400ns	0.125663706	0.19739556	36.33914246
75000	6.2	9	0.688888889	0.991227901	30.50146304	-3.237016399	-0.076529642	200ns	0.09424778	0.132551532	28.89725379
100000	7.2	10.8	0.666666667	0.99503719	33.00082919	-3.521825181	-0.043213738	200ns	0.125663706	0.099668652	26.08147397
125000	5.8	9.4	0.617021277	0.996815279	38.10074044	-4.193997201	-0.027706281	100ns	0.078539816	0.079829986	1.616146315
150000	5.6	9.4	0.595744681	0.997785158	40.29329098	-4.498796532	-0.01925921	70ns	0.065973446	0.066568164	0.893397109
200000	5.8	9.4	0.617021277	0.998752339	38.22079283	-4.193997201	-0.010843813	40ns	0.050265482	0.049958396	0.614684941
500000	6.8	9.4	0.723404255	0.99980006	27.64510783	-2.812378818	-0.001736831	30ns	0.09424778	0.019997334	371.3017232
1000000	6.8	9.4	0.723404255	0.999950004	27.65595754	-2.812378818	-0.000434273	0	0	0.009999667	100

Note

Measurements could not be taken for frequencies below 10 Hz, as the output signal was very small and largely dominated by noise.

Reasons for Deviations

1. The resistance used in the circuit was comparable to the input resistance of the oscilloscope probe (10 MΩ)
2. The capacitance value (10 pF) was very small, making the circuit highly sensitive to parasitic effects (even a human hand near the breadboard effected the output).

Experimental Cut-off Frequency and Roll-off Slope

Cut-off Frequency: The cut-off frequency is defined as the frequency at which the gain is -3 dB. From the data:

$$20 \log |H| \big|_{f=17 \text{ kHz}} \approx -3.4 \text{ dB}$$

$$f_c \approx 17 \text{ kHz}$$

Roll-off Slope: The slope is obtained from points of straight-line like region

$$\text{Slope} = \frac{\Delta(20 \log |H|)}{\Delta \log_{10} f}$$

Taking values at $f_1 = 300 \text{ Hz}$ (-33.50 dB) and $f_2 = 2500 \text{ Hz}$ (-13.28 dB):

$$\Delta(20 \log |H|) = (-13.28) - (-33.50) = 20.21 \text{ dB}$$

$$\Delta \log_{10} f = \log_{10} \left(\frac{2500}{300} \right) = \log_{10}(8.33) \approx 0.921$$

$$\therefore \text{Slope} \approx \frac{20.21}{0.921} \approx 21.95 \text{ dB/decade}$$

Parameter	Experimental Value	Theoretical Value
Cut-off Frequency, f_c	$\approx 17 \text{ kHz}$	10 kHz
Roll-off Slope	$\approx 21 \text{ dB/decade}$	20 dB/decade

Simulation using SPICE

We have simulated the filter using ngspice and plotted the output using gnuplot.

Credits to **Sarthak** for help with the NGSpice simulation code throughout the entire experiment.

```
*Netlist
Vin in 0 AC 10
C in out 15p
R out 0 1Meg

.control
AC DEC 30 10 1meg
set wr_singlescale
wrdata stage1.dat 20*log10(abs(v(out)/v(in))) phase(v(out))
.endc
.end
```

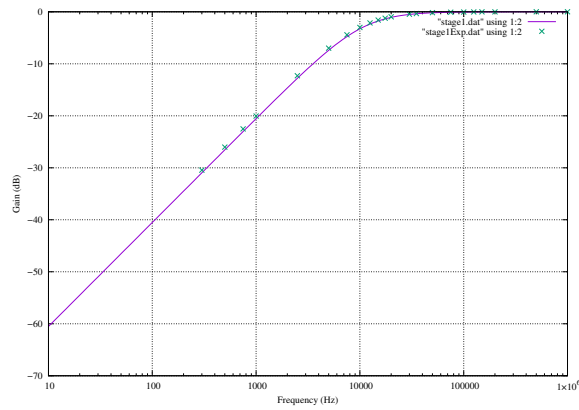
```
set terminal postscript eps enhanced color font 'Times,12'
set logscale x
set output 'stage1.eps'
set xlabel 'Frequency (Hz)'
set ylabel 'Gain (dB)'
set grid
plot "stage1.dat" using 1:2 with lines, "stage1Exp.dat" using 1:2
with points pt 2
set output
```

```
set terminal postscript eps enhanced color font 'Times,12'
set logscale x
set output 'stage1phExp.eps'

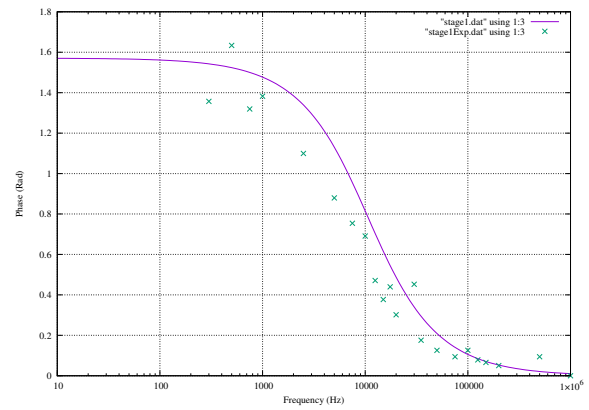
set xlabel 'Frequency (Hz)'
set ylabel 'Phase (Rad)'
set grid

plot "stage1.dat" using 1:3 with lines, "stage1Exp.dat" using 1:3
with points pt 2

set output
```



Magnitude response of the single-stage RC high-pass filter

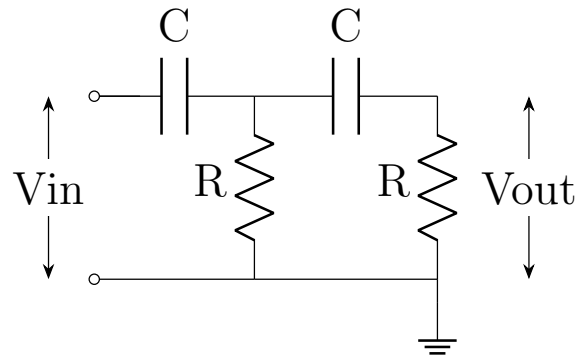


Phase response of the single-stage RC high-pass filter

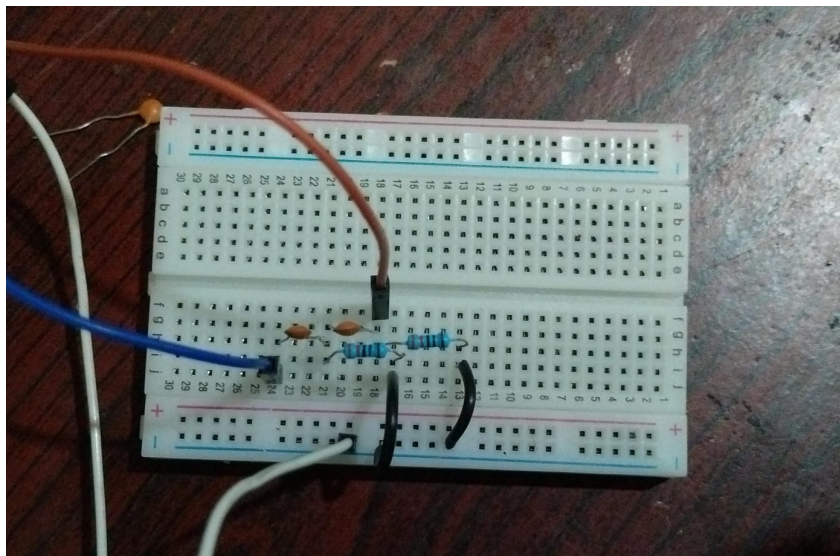
Stage 2 : Circuit Design and Setup

The circuit is obtained by cascading two single stage RC high-pass filters with $R = 1\text{ M}\Omega$ and $C = 15\text{ pF}$. The theoretical cut-off frequency is given by

$$f_c = \frac{1}{2\pi RC} \approx 10.6\text{ kHz.}$$



Breadboard Implementation



Breadboard implementation of the double-stage filter.

Observations

Freq(num)	Vout	Vin	Gain	Gain(theo)	Gain error(%)	20log—H—	20log—H'theo—	delT	delPhase	delPhase(Theo)	Phase error(%)
10	0	8.8	0	1.00E-06	100	#NUM!	-120.0000087		#VALUE!	3.139592654	#VALUE!
50	0	9.2	0	2.50E-05	100	#NUM!	-92.04141697		#VALUE!	3.131592737	#VALUE!
100	0.008	8.6	0.000930233	1.00E-04	830.3255814	-60.62816929	-80.00086855		#VALUE!	3.12159332	#VALUE!
300	0.008	8.6	0.000930233	0.000899191	3.452196382	-60.62816929	-60.9229636		#VALUE!	3.081610644	#VALUE!
500	0.008	9.8	0.000816327	0.002493766	67.26530612	-61.76272177	-52.06288745		#VALUE!	3.041675862	#VALUE!
750	0.048	9.8	0.004897959	0.005593536	12.43537415	-46.19969677	-45.04627069	780us	3.675663405	2.991872958	22.85492921
1000	0.048	9.6	0.005	0.00990099	49.5	-46.02059991	-40.08642748	460us	2.890265241	2.942255349	1.767015474
2500	0.68	9.6	0.070833333	0.058823529	20.41666667	-22.99524641	-24.60897843	150us	2.35619449	2.651635327	11.14183516
5000	1.2	9.6	0.125	0.2	37.5	-18.06179974	-13.97940009	57us	1.790707813	2.214297436	19.12975268
7500	1.84	9.6	0.191666667	0.36	46.75925926	-14.3490682	-8.873949985	31us	1.460840584	1.854590436	21.23109472
10000	2.8	9.4	0.29787234	0.5	40.42553191	-10.51939645	-6.020599913	20us	1.256637061	1.570796327	20
12500	4.8	9.6	0.5	0.609756098	18	-6.020599913	-4.296876961	14us	1.099557429	1.349481884	18.52003043
15000	3.6	9.6	0.375	0.692307692	45.83333333	-8.519374645	-3.194016857	9us	0.848230016	1.176005207	27.87191661
17500	4	9.6	0.416666667	0.753846154	44.72789116	-7.604224834	-2.454345532	8us	0.879645943	1.038292228	15.27954088
20000	4.4	9.4	0.468085106	0.8	41.4893617	-6.593503542	-1.93820026	6us	0.753982237	0.927295218	18.69016229
30000	6.4	9.4	0.680851064	0.9	24.3498818	-3.338957592	-0.915149811	4us	0.753982237	0.643501109	17.16875489
35000	5.2	9.4	0.553191489	0.924528302	40.16500217	-5.142490199	-0.681595791	2.4us	0.527787566	0.556599318	5.176390138
50000	5.2	9.2	0.565217391	0.961538462	41.2173913	-4.955689674	-0.340666786	1.56us	0.490088454	0.39479112	24.1386722
75000	4.08	9.4	0.434042553	0.982532751	55.82411348	-7.24935381	-0.153059285	500ns	0.235619449	0.265103065	11.12156724
100000	7.2	9.4	0.765957447	0.99009901	22.63829787	-2.315907143	-0.086427476	300ns	0.188495559	0.199337305	5.438894525
125000	6	9.44	0.63559322	0.9936407	36.03389831	-3.936414878	-0.055412562	680ns	0.534070751	0.159659971	234.5051025
150000	5.2	9.44	0.550847458	0.995575221	44.67043315	-5.179373013	-0.038518421	160ns	0.150796447	0.133136328	13.26468902
200000	5.6	9.44	0.593220339	0.997506234	40.52966102	-4.535679346	-0.021687626	90ns	0.113097336	0.099916791	13.19152056
500000	5	9.52	0.525210084	0.99960016	47.45798319	-5.593338881	-0.003473661	12ns	0.037699112	0.039994668	5.739655362
1000000	5	9.36	0.534188034	0.99990001	46.5758547	-5.446116888	-0.000868546	4ns	0.025132741	0.019999333	25.66789482

Reasons for Deviations

1. The resistance used in the circuit was comparable to the input resistance of the oscilloscope probe (10 M Ω)
2. The capacitance value (10 pF) was very small, making the circuit highly sensitive to parasitic effects (even a human hand near the breadboard effected the output).
3. Loading between the first and second stage caused major deviation.
4. The overall transfer function is the product of two single-stage responses, so small errors in each stage get added, leading to larger deviation.

Experimental Cut-off Frequency and Roll-off Slope

Cut-off Frequency: The cut-off frequency is defined as the frequency at which the gain drops to -6 dB. From the data:

$$20 \log |H| \Big|_{f=12.5 \text{ kHz}} \approx -6 \text{ dB}$$

This is close to the theoretical -6 dB point. Hence, the experimentally obtained cut-off frequency is:

$$f_c \approx 12.5 \text{ kHz}$$

Roll-off Slope The slope is obtained from the linear stop-band region using

$$\text{Slope} = \frac{\Delta(20 \log |H|)}{\Delta \log_{10} f}$$

Taking values at $f_1 = 500 \text{ Hz}$ (-61.76 dB) and $f_2 = 2500 \text{ Hz}$ (-22.99 dB):

$$\Delta(20 \log |H|) = (-22.99) - (-61.76) = 38.77 \text{ dB}$$

$$\Delta \log_{10} f = \log_{10}\left(\frac{2500}{500}\right) = \log_{10}(5) \approx 0.699$$

$$\therefore \text{Slope} \approx \frac{38.77}{0.699} \approx 55.47 \text{ dB/decade}$$

Parameter	Experimental Value	Theoretical Value
Cut-off Frequency, f_c	$\approx 12.5 \text{ kHz}$	10 kHz
Roll-off Slope	$\approx 55 \text{ dB/decade}$	40 dB/decade

Simulation using SPICE

We have simulated the filter using ngspice and plotted the output using gnuplot.

```
*Netlist
Vin in 0 AC 10
C1 in x 15p
R1 x 0 10meg
C2 x out 15p
R2 out 0 1meg

.control
AC DEC 30 10 1meg
set wr_singlescale
wrdata stage2.dat 20*log10(abs(v(out)/v(in))) phase(v(out))
.endc
.end
```

```
set terminal postscript eps enhanced color font 'Times,12'
set logscale x
set output 'stage2.eps'

set xlabel 'Frequency (Hz)'
set ylabel 'Gain (dB)'
set grid

plot "stage2.dat" using 1:2 with lines, "stage2Exp.dat" using 1:2
with points pt 2

set output
```

```
set terminal postscript eps enhanced color font 'Times,12'
set logscale x
set output 'stage2ph.eps'

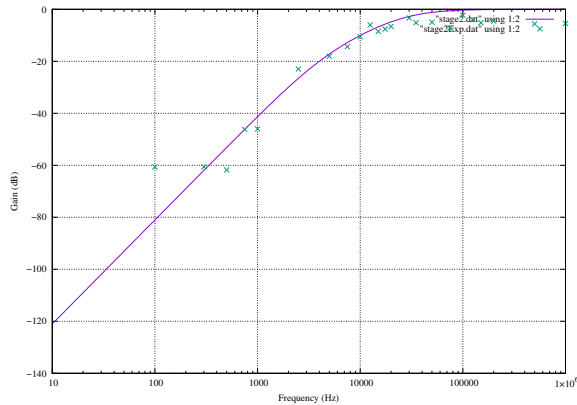
set xlabel 'Frequency (Hz)'
set ylabel 'Phase (Rad)'
set grid
```

```

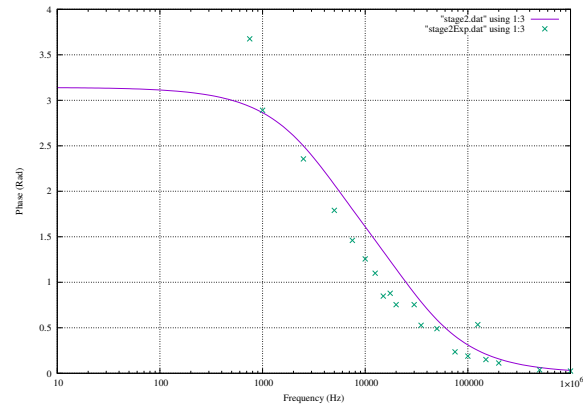
plot "stage2.dat" using 1:3 with lines, "stage2Exp.dat" using 1:3
    with points pt 2

set output

```



Magnitude response of the double-stage RC high-pass filter



Phase response of the double-stage RC high-pass filter

Closest fit for experimental data : Stage 1 and Stage 2

The experimental gain and phase responses for both single-stage and double-stage RC high-pass filters were compared against simulation models. Mean squared error (MSE) minimization technique was used to find the optimal RC value that minimized the mean squared error.

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
pi = np.pi
# Read Stage 1 and Stage 2 data
df1 = pd.read_csv("Task1_EE2501 - Stage 1.csv")
df2 = pd.read_csv("Task1_EE2501 - Stage 2.csv")

# Extract columns for Stage 1
freqs1 = df1['Freq(num)']
H_exp1 = df1['Gain']
H_theo1 = df1['Gain(theo)']
phase_exp1 = df1['delPhase']
phase_theo1 = df1['delPhase(Theo)']

# Extract columns for Stage 2
freqs2 = df2['Freq(num)']
H_exp2 = df2['Gain']
H_theo2 = df2['Gain(theo)']
phase_exp2 = df2['delPhase']
phase_theo2 = df2['delPhase(Theo)']
# Mean squared error between experimental and theoretical data

```

```

def calculateMSE(expe, theo):
    if len(expe) != len(theo):
        raise ValueError("Length mismatch")
    return np.mean((np.array(expe) - np.array(theo))**2)

# Stage 1 magnitude and phase
def H_stage1(fs, RC):
    omegas = 2*pi*fs
    return (omegas*RC) / np.sqrt(1+(omegas*RC)**2)

def phase_stage1(fs, RC):
    omegas = 2*pi*fs
    return np.arctan(1/(omegas*RC))

# Stage 2 magnitude and phase
def H_stage2(fs, RC):
    omegas = 2*pi*fs
    num = (omegas*RC)**2
    den = np.sqrt((1-(omegas*RC)**2)**2 + (3*omegas*RC)**2)
    return num/den

def phase_stage2(fs, RC):
    omegas = 2*pi*fs
    num = (RC*omegas)**2 - 1
    den = 3*RC*omegas
    return pi/2 - np.arctan(num/den)

# Fit RC by minimizing MSE
def curve_fit(freqs, H_exp, RC_init, step, H_stage):
    RCs = []
    MSEs = []

    # Sweep RC scaling factor from 'step' to 100 RC_init ,
    # compute MSE for each
    for i in np.arange(step, 100, step):
        RC = RC_init * i
        H_RC_theo = H_stage(freqs, RC)
        RCs.append(RC)
        MSEs.append(calculateMSE(H_exp, H_RC_theo))

    # Find RC corresponding to the minimum MSE
    min_MSE_index = np.argmin(MSEs)
    print(f"RC corresponding to minimum MSE is {RCs[min_MSE_index]} which is {step * (min_MSE_index + 1)} times of initial RC.")
    print(f"fc corresponding to minimum MSE is {1/(2*pi*RCs[min_MSE_index])}")
    return RCs[min_MSE_index]

# Plot magnitude and phase
def plot_curves(freqs, H_exp, H_theo, phase_exp, phase_theo, min_RC, title, H_stage, phase_stage):

```

```

# Plotting Frequency response with experimental, theoretical
# with initial RC and RC with minimum MSE
plt.figure(figsize=(14,6))
plt.subplot(1, 2, 1)
plt.semilogx(freqs, 20*np.log10(H_exp), label="Experimental")
plt.semilogx(freqs, 20*np.log10(H_theo), label="Theoretical")
plt.semilogx(freqs, 20*np.log10(H_stage(freqs, min_RC)),
    label="Theoretical with min. MSE")
plt.xlabel("Frequencies (Hz)")
plt.ylabel("20log|H|")
plt.title(f"{title} - Frequency Response")
plt.legend()
plt.grid()

# Plotting Phase with experimental, theoretical with initial
# RC and RC with minimum MSE
plt.subplot(1, 2, 2)
plt.semilogx(freqs, phase_exp, label="Experimental")
plt.semilogx(freqs, phase_theo, label="Theoretical")
plt.semilogx(freqs, phase_stage(freqs, min_RC), label="
    Theoretical with min. MSE")
plt.xlabel("Frequencies (Hz)")
plt.ylabel("Phase")
plt.title(f"{title} - Phase Plot")
plt.legend()
plt.grid()
plt.show()

# Circuit parameters: R = 1 M , C = 15 pF
# For stage 1
print("For stage1:")
min_RC_stage1 = curve_fit(freqs1, H_exp1, 1e6*15e-12, 1e-3,
    H_stage1)

# For stage 2
print("For stage2:")
min_RC_stage2 = curve_fit(freqs2, H_exp2, 1e6 * 15e-12, 1e-3,
    H_stage2)
# Plotting frequency response and phase plot of stage 1 and stage
# 2
plot_curves(freqs1, H_exp1, H_theo1, phase_exp1, phase_theo1,
    min_RC_stage1, "Stage 1", H_stage1, phase_stage1)
plot_curves(freqs2, H_exp2, H_theo2, phase_exp2, phase_theo2,
    min_RC_stage2, "Stage 2", H_stage2, phase_stage2)

```

By running the above program, we obtained the best-fit values of RC and the corresponding cutoff frequencies f_c for both stages:

- Stage 1:

$$RC_{\min \text{ MSE}} = 9.435 \times 10^{-6} \text{ s} \quad (0.629 \times \text{initial } RC)$$

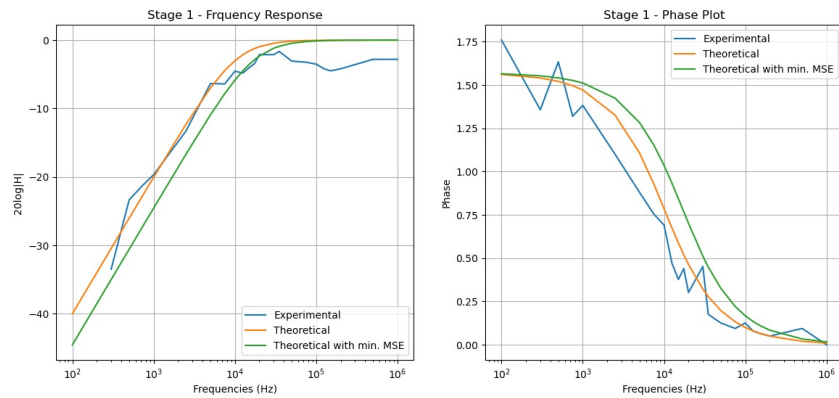
$$f_{c, \min \text{ MSE}} = 16868.57 \text{ Hz}$$

- Stage 2:

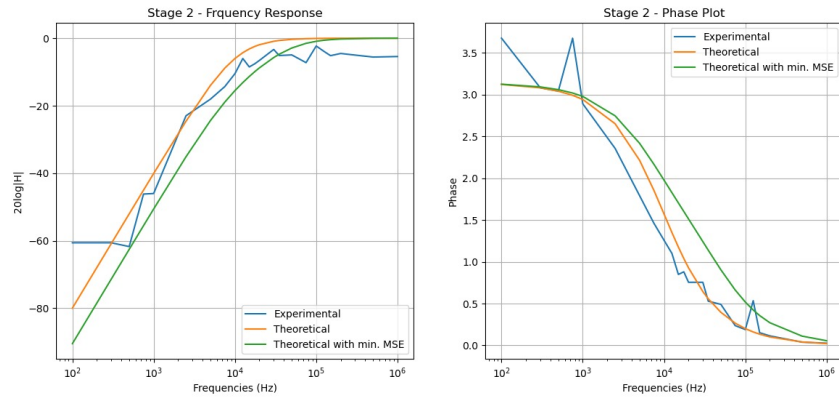
$$RC_{\min \text{ MSE}} = 8.685 \times 10^{-6} \text{ s} \quad (0.579 \times \text{initial } RC)$$

$$f_{c, \min \text{ MSE}} = 18325.27 \text{ Hz}$$

The frequency response (gain) and phase plots for the best-fit, experimental, and theoretical data are shown below for both Stage 1 and Stage 2.

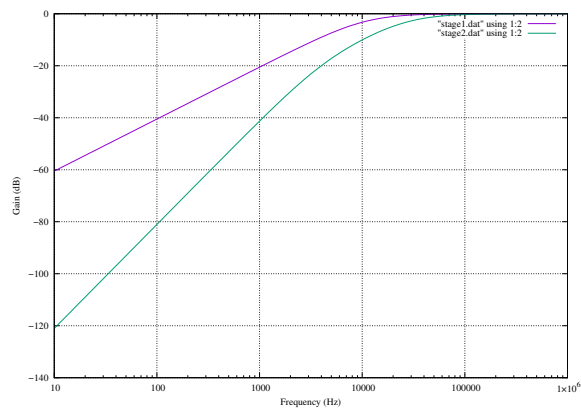


Stage 1: Experimental, Theoretical, and Best-fit frequency response and phase plots.

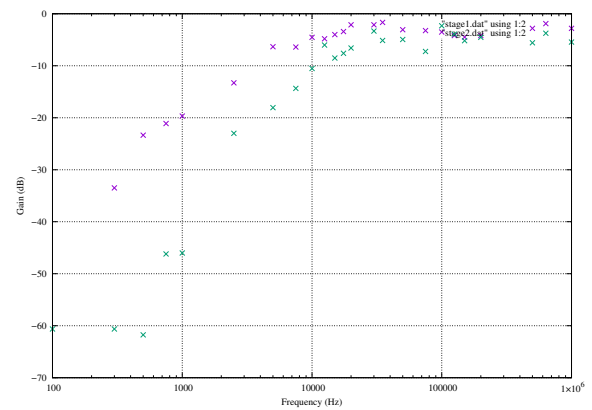


Stage 2: Experimental, Theoretical, and Best-fit frequency response and phase plots.

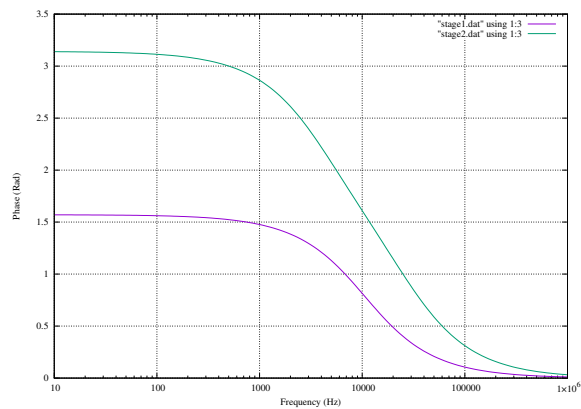
Stage-1 v/s Stage-2 Plots



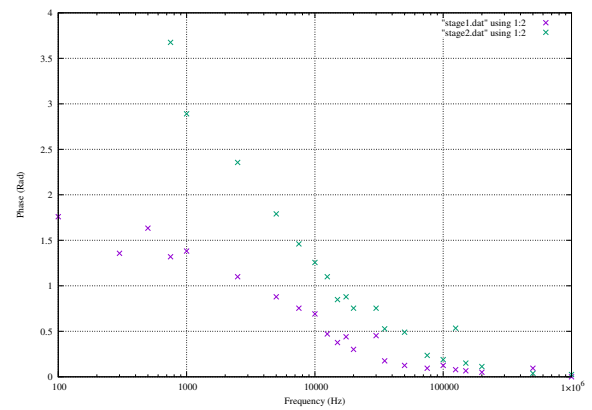
Theoretical Gain of Stage 1 and Stage 2



Experimental Gain of Stage 1 and Stage 2



Theoretical Phase of Stage 1 and Stage 2



Experimental Phase of Stage 1 and Stage 2

Task 2 : Improvement in double stage filter

Analysis of Loading in Double-Stage Filter

When two isolated filter are cascaded ideally, the overall transfer function is simply the product of their individual transfer functions. This is because the output of the first stage, V_{out1} , cancels with the input of the second stage, V_{in2} , resulting in

$$H_{overall}(j\omega) = H_1(j\omega) \cdot H_2(j\omega).$$

However, in the practical case of a double-stage RC filter, the output of the first stage is influenced by the input impedance of the second stage. As a result, $H = H_1 \cdot H_2$ relationship does not hold. Instead, the overall transfer function must be derived by directly applying Kirchhoff's Current Law (KCL) in the overall circuit.

We used SymPy in Python to set up and solve the governing equations for the node voltages.

Finally, by comparing the simplified forms of $H(j\omega)$ and $H_1(j\omega)$, it was observed that they differ, thus confirming the loading effect.

```
from sympy import symbols, Eq, solve, pprint, I
import numpy as np

# Define symbols
x, y, R, w, C, R2, C2, V = symbols('x y R w C R2 C2 V')

# Equations based on KCL
eq1 = Eq(x*(I*w*C + 1/R + I*w*C2) + y*(-I*w*C2), I*w*C*V)
eq2 = Eq(x*(I*w*C2) + y*(-1/R2 - I*w*C2), 0)

# Solve for node voltages
solution = solve((eq1, eq2), (x, y))

# Transfer function H = Vout/Vin
H = solution[y] / V

# Expected transfer function without loading
H1 = R**2 / (R + 1/(I*w*C))**2

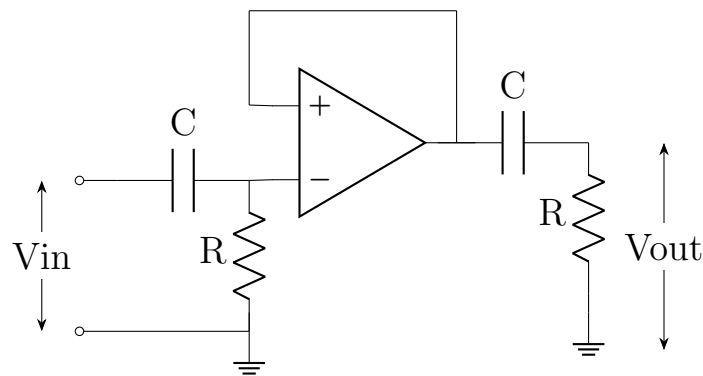
# Simplify results
H_simplified = H.simplify()
H1_simplified = H1.simplify()
```

Active Scheme Design

In order to avoid the loading effect we introduce a buffer between the two stages. The buffer is implemented using a unity-gain op-amp. The role of the buffer is to present a very high input impedance to the first stage and a very low output impedance to the second stage. As a result, the output of the first filter stage is not affected by the loading due to the second stage. Effectively, the output of the first stage becomes the direct input to the second stage.

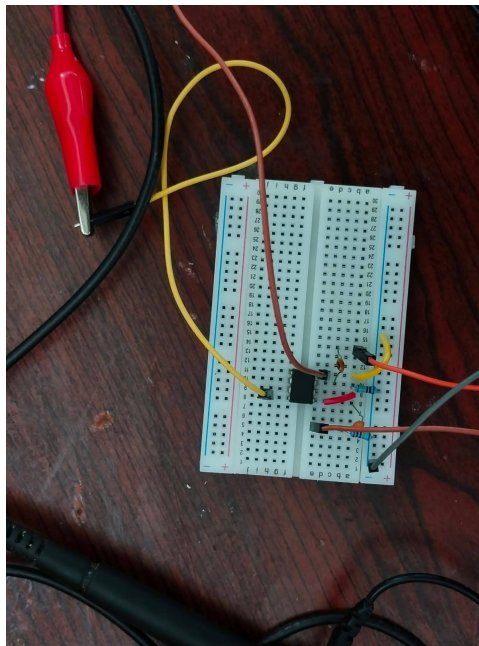
This ensures that the overall transfer function of the double-stage filter is the product of the individual transfer functions:

$$H_{\text{overall}}(j\omega) = H_1(j\omega) \cdot H_2(j\omega).$$



Breadboard Implementation

The physical implementation of the circuit on breadboard is shown below



Breadboard implementation of the 2nd stage active scheme

Observations

Freq	Vout	Vin	Gain	H(theo)	H error(%)	20log—H—	20log—H—theo—	delT	delPhase	delPhase(Theo)	Phase error(%)
10	0	1.16	0	1.00E-06	100	#NUM!	-120.0000087		#VALUE!	3.139592654	#VALUE!
50	0	1	0	2.50E-05	100	#NUM!	-92.04141697		#VALUE!	3.131592737	#VALUE!
100	0	1	0	1.00E-04	100	#NUM!	-80.00086855		#VALUE!	3.12159332	#VALUE!
300	0	1	0	0.000899191	100	#NUM!	-60.9229636		#VALUE!	3.081610644	#VALUE!
500	0	1	0	0.002493766	100	#NUM!	-52.06288745		#VALUE!	3.041675862	#VALUE!
750	0.008	0.912	0.00877193	0.005593536	56.82261209	-41.13809703	-45.04627069		#VALUE!	2.991872958	#VALUE!
1000	0.008	0.904	0.008849558	0.00990099	10.61946903	-41.06156887	-40.08642748		#VALUE!	2.942255349	#VALUE!
2500	0.032	0.912	0.035087719	0.058823529	40.35087719	-29.0968972	-24.60897843	230us	3.612831552	2.651635327	36.24918609
5000	0.136	0.888	0.153153153	0.2	23.42342342	-16.29748115	-13.97940009	60us	1.884955592	2.214297436	14.87342387
7500	0.368	0.864	0.425925926	0.36	18.3127572	-7.413318476	-8.873949985	30us	1.413716694	1.854590436	23.77202715
10000	0.464	0.864	0.537037037	0.5	7.407407407	-5.399915238	-6.020599913	25us	1.570796327	1.570796327	0
12500	0.488	0.864	0.564814815	0.609756098	7.37037037	-4.96187841	-4.296876961	15us	1.178097245	1.349481884	12.70003261
15000	0.52	0.896	0.580357143	0.692307692	16.17063492	-4.726093321	-3.194016857	8.8us	0.829380461	1.176005207	29.47476291
17500	0.544	1.06	0.513207547	0.753846154	31.92144782	-5.794139311	-2.454345532	6.8us	0.747699052	1.038292228	27.98760975
20000	0.632	0.848	0.745283019	0.8	6.839622642	-2.553575479	-1.93820026	5.2us	0.653451272	0.927295218	29.53147398
30000	0.608	0.88	0.690909091	0.9	23.23232323	-3.211581858	-0.915149811	2.8us	0.527787566	0.643501109	17.98187158
35000	0.608	1.03	0.590291262	0.924528302	36.15216961	-4.578672909	-0.681595791	2.8us	0.61575216	0.565599318	10.62754484
50000	0.624	0.832	0.75	0.961538462	22	-2.498774732	-0.340666786	1.2us	0.376991118	0.39479112	4.508713692
75000	0.648	1	0.648	0.982532751	34.048	-3.768499883	-0.153059285	300ns	0.141371669	0.265103065	46.67294034
100000	0.616	0.864	0.712962963	0.99009901	27.99074074	-2.938606060	-0.086427476	280ns	0.175929189	0.199337305	11.74296822
125000	0.648	0.872	0.743119266	0.9936407	25.21247706	-2.578829581	-0.055412562	100ns	0.078539816	0.159659971	50.80807316
150000	0.568	0.936	0.606837607	0.995575221	39.04653371	-4.338550261	-0.038518421	100ns	0.09424778	0.133136328	29.20956936
200000	0.648	0.888	0.72972973	0.997506234	26.84459459	-2.736759198	-0.021687626	70ns	0.087964594	0.099916791	11.96215068
500000	0.512	0.88	0.581818182	0.99960016	41.79490909	-4.704254223	-0.003473661	-320ns	-1.005309649	0.039994668	2413.60919
1000000	0.312	0.856	0.364485981	0.99990001	63.54775701	-8.766383413	-0.000868546	-270ns	-1.696460033	0.019999333	8382.5829

Note

Deviations might have arise due to finite gain and input/output impedance effects of the op-amp

Simulation using SPICE

We have simulated the filter using ngspice and plotted the output using gnuplot.

```
*Netlist
Vcc vcc 0 5
Vee vee 0 -5
Vin in 0 AC 1
C1 in 1 15p
R1 1 out 1Meg
C2 1 2 15p
R2 2 0 1Meg
XU1 2 out vcc vee out LM358/NS
.include LM358.mod

.control
AC DEC 30 10 1meg
set wr_singlescale
wrdata active.dat 20*log10(abs(v(out)/v(in))) phase(v(out))
.endc
.end
```

```
set terminal postscript eps enhanced color font 'Times,12'
set logscale x
set output 'activeExp.eps'

set xlabel 'Frequency (Hz)'
set ylabel 'Gain (dB)'
set grid
```

```
plot "active.dat" using 1:2 with lines, "activeExp.dat" using 1:2
    with points pt 2
```

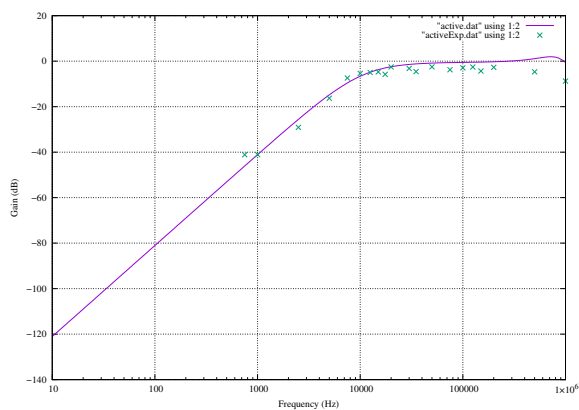
```
set output
```

```
set terminal postscript eps enhanced color font 'Times,12'
set logscale x
set output 'activephExp.eps'
```

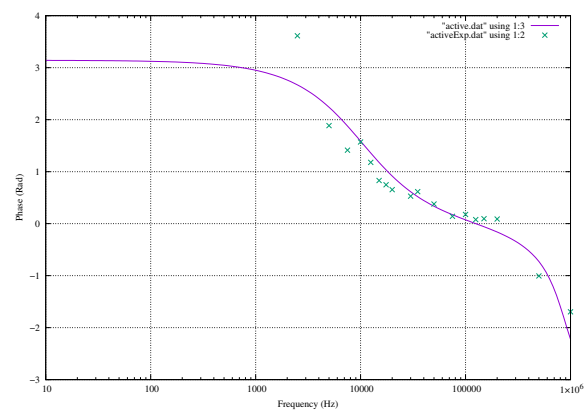
```
set xlabel 'Frequency (Hz)'
set ylabel 'Phase (Rad)'
set grid
```

```
plot "active.dat" using 1:3 with lines, "activeExp.dat" using 1:3
    with points pt 2
```

```
set output
```



Magnitude response of the double-stage active filter



Phase response of the double-stage active filter

Note

We imported LM-358 model for SPICE from [here](#)

Passive Scheme Design

Passive improvement was done by analyzing the transfer function of circuit. While solving for H , the second-stage resistor and capacitor were kept the same, while the first-stage components were varied to observe their effect on H

From the equation, it was clear that to improve the response of the double-stage filter, the resistance of 1st stage needed to be decreased and the capacitance needed to be increased. The values of R and C were adjusted by a factor of 1500 based on the available component values.

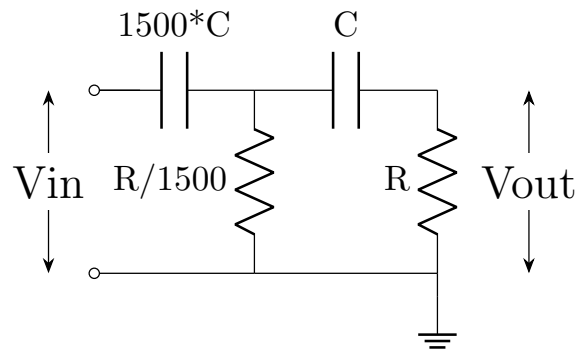
```
from sympy import symbols, Eq, solve, I

x, y, R, w, C, R2, C2, V = symbols('x y R w C R2 C2 V')

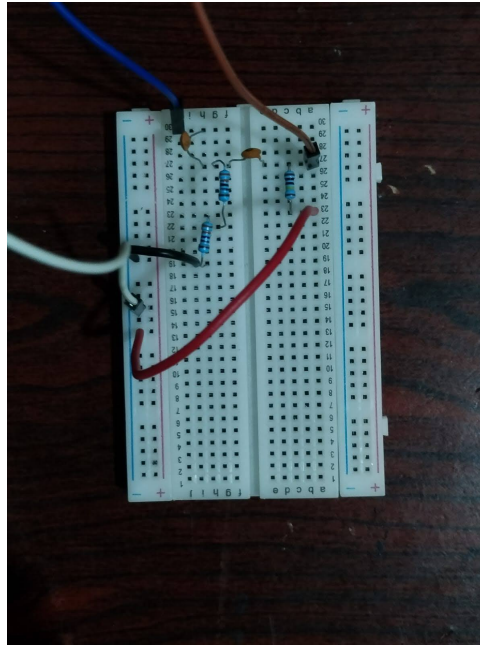
eq1 = Eq(x*(I*w*C + 1/R + I*w*C2) + y*(-I*w*C2), I*w*C*V)
eq2 = Eq(x*(I*w*C2) + y*(-1/R2 - I*w*C2), 0)

solution = solve((eq1, eq2), (x, y))
H = solution[y] / V

H_subs = H.subs({R2: 1e6, C2: 15e-12})
H_simplified = H_subs.simplify()
```



Breadboard Implementation



Breadboard implementation of the 2nd stage passive scheme

Observations

Freq	Vout	Vin	H	H(theo)	H error(%)	20log—H—	20log—H theo—	delT	delPhase	delPhase(Theo)	Phase error(%)
10	0	5.12	0	0.000001074795876	100.00	#NUM!	-119.3734802		0	3.138482484	100
50	0	5.12	0	0.00002686747134	100.00	#NUM!	-91.41546411		0	3.126042606	100
100	0	5.12	0	0.0001074395823	100.00	#NUM!	-79.37671378		0	3.110497569	100
250	0	5.12	0	0.0006701759793	100.00	#NUM!	-63.47622285		0	3.063942412	100
500	0	5.12	0	0.002662071457	100.00	#NUM!	-51.49560582		0	2.986910434	100
750	0.096	5.12	0.01875	0.005921639305	216.64	-34.53997456	-44.55116099	630us	2.968805058	2.911081993	1.982873207
1000	0.112	5.12	0.021875	0.01036459651	111.06	-33.20103877	-39.68895201	480us	3.015928947	2.836980666	6.30770185
2500	0.34	5.04	0.06746031746	0.05531593897	21.95	-23.41903239	-25.14299422	160us	2.513274123	2.446734177	2.719541295
5000	0.68	4.96	0.1370967742	0.1563610546	12.32	-17.25945528	-16.11742818	68us	2.136283004	2.010372344	6.263051737
6000	0.86	4.96	0.1733870968	0.196868557	11.98	-15.2196645	-14.11125504	52us	1.960353816	1.888217773	3.820324332
7000	1	4.88	0.2049180328	0.236380178	13.31	-13.76839644	-12.52777889	40us	1.759291886	1.784883447	1.433794529
8000	1.12	4.96	0.2258064516	0.2743102011	17.68	-12.92527308	-11.23516083	33us	1.658760921	1.6955907	2.172091346
9000	1.24	4.88	0.2540983607	0.3106858135	18.21	-11.89996274	-10.15357154	29us	1.639911365	1.616995909	1.417162262
10000	1.36	4.96	0.2741935484	0.345475294	20.63	-11.23885536	-9.231660098	25us	1.570796327	1.546750908	1.554576043
12500	1.64	4.88	0.33606055738	0.4255207475	21.02	-9.471519479	-7.421585195	18us	1.413716694	1.397794329	1.139106442
15000	1.88	4.8	0.3916666667	0.495950413	21.03	-8.141667762	-6.091234874	14us	1.319468915	1.275662238	3.434034156
17500	2.16	4.72	0.4576271186	0.5573699856	17.90	-6.78976495	-5.077128435	11us	1.209513172	1.172294858	3.174825272
20000	2.32	4.72	0.4915254237	0.6105837425	19.50	-6.169080275	-4.285095269	9us	1.130973355	1.083135028	4.416654042
21000	2.4	4.72	0.5084745763	0.6297726828	19.26	-5.874615138	-4.016323627	8.8us	1.161132645	1.050767058	10.50333523
22000	2.4	4.72	0.5084745763	0.6478524697	21.51	-5.874615138	-3.770477626	8us	1.105840614	1.020060835	8.409280765
23000	2.56	4.72	0.5423728814	0.6648820793	18.43	-5.314040666	-3.54510745	7.6us	1.098300792	0.9908936189	10.83942522
24000	2.56	4.72	0.5423728814	0.6809195015	20.35	-5.314040666	-3.338084549	4.2us	0.633345079	0.9631564323	34.24276081
25000	2.64	4.64	0.5689655172	0.6960212779	18.25	-4.898281074	-3.147549669	6us	0.9424777961	0.9367516642	0.611275334
26000	2.72	4.64	0.5862068966	0.7102421415	17.46	-4.63898153	-2.971871257	6us	0.9801769079	0.9115912357	7.523730981
27000	2.72	4.72	0.5762711864	0.7236347482	20.36	-4.787461892	-2.809611739	5.6us	0.9500176184	0.8875951701	7.032761151
28000	2.72	4.64	0.5862068966	0.7362494885	20.38	-4.63898153	-2.659499879	5.2us	0.9148317807	0.8646904659	5.798758842
29000	2.8	4.64	0.6034482759	0.7481343664	19.34	-4.387198984	-2.5204079	5.2us	0.9475043443	0.8428101949	12.42203169
30000	2.96	4.64	0.6379310345	0.7593349351	15.99	-3.90452539	-2.391332377	4.8us	0.9047786842	0.8218927719	10.08475986
35000	3.12	4.56	0.6842105263	0.8064964488	15.16	-3.296204973	-1.867950811	3.6us	0.7916813487	0.7298988633	8.464526874
40000	3.2	4.56	0.701754386	0.8420223523	16.66	-3.076297287	-1.493527592	3.2us	0.8042477193	0.6549827251	22.78914977
45000	3.28	4.48	0.7321428571	0.8691602193	15.76	-2.708083406	-1.218003183	2.4us	0.6785840132	0.5930752256	14.41786538
50000	3.36	4.48	0.75	0.8902003525	15.75	-2.498774732	-1.010244761	2us	0.6283185307	0.54124008	16.08869224
75000	3.6	4.48	0.8035714286	0.9465787087	15.11	-1.899510265	-0.4768653663	1us	0.471238898	0.3738676341	26.04431489
100000	3.76	4.4	0.8545283019	0.9689053862	11.80	-1.365296631	-0.2743725968	600ns	0.3769911184	0.2841844375	32.65720029
250000	3.84	4.32	0.8888888889	0.994829007	10.65	-1.023050449	-0.04503120331	80ns	0.1256637061	0.1154065182	8.887875717
500000	3.92	4.32	0.9074074074	0.9986999025	9.14	-0.8439535959	-0.01129985034	40ns	0.1256637061	0.05783151294	117.2927868
750000	3.92	4.32	0.9074074074	0.9994215695	9.21	-0.8439535959	-0.005025637449	20ms	0.09424777961	0.03857026147	144.353489
1000000	4	4.24	0.9433962264	0.9996745126	5.63	-0.5061173053	-0.002827607406	10ms	0.06283185307	0.02893187975	117.171693
2000000	3.92	4.24	0.9245283019	0.9999185992	7.54	-0.6815957914	-0.0007070674162	4ns	0.05026548246	0.01446795817	247.4262357
2500000	3.84	4.24	0.9056603774	0.9999479012	9.43	-0.8606926445	-0.0004525358663	0	0	0.01157456035	100
5000000	3.76	4.08	0.9215686275	0.9999869746	7.84	-0.7094463632	-0.0001131382069	0	0	0.005787409399	100
7500000	3.76	4	0.94	0.9999942109	6.00	-0.537442928	-0.00005028399654	0	0	0.003858288887	100
10000000	3.6	3.92	0.9183673469	0.9999967436	8.16	-0.7396713251	-0.00002828481677	-1ns	-0.06283185307	0.002893720853	2271.317009

Simulation using SPICE

We have simulated the filter using ngspice and plotted the output using gnuplot.

```
*Netlist
Vin in 0 AC 5
C1 in x 25n
R1 x 0 660
C2 x out 15p
R2 out 0 1meg

.control
AC DEC 30 10 10meg
set wr_singlescale
wrdata passive.dat 20*log10(abs(v(out)/v(in))) phase(v(out))
.endc
.end
```

```
set terminal postscript eps enhanced color font 'Times,12'
set logscale x
set output 'passiveExp.eps'

set xlabel 'Frequency (Hz)'
set ylabel 'Gain (dB)'
set grid

plot "passive.dat" using 1:2 with lines, "passiveExp.dat" using
    1:2 with points pt 2

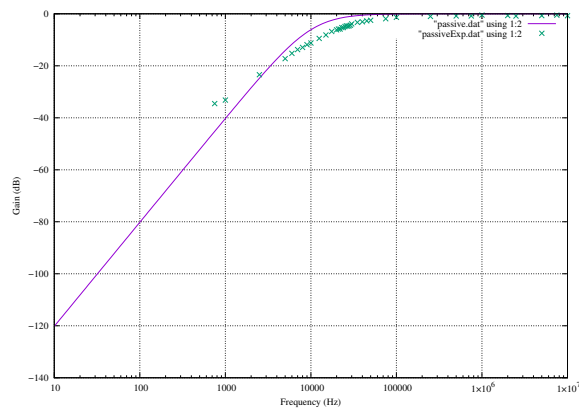
set output
```

```
set terminal postscript eps enhanced color font 'Times,12'
set logscale x
set output 'passivephExp.eps'

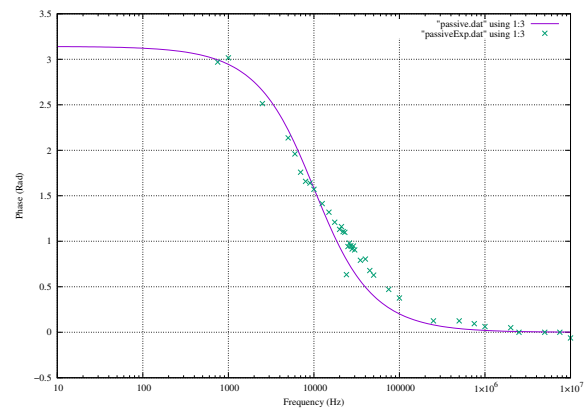
set xlabel 'Frequency (Hz)'
set ylabel 'Phase (Rad)'
set grid

plot "passive.dat" using 1:3 with lines, "passiveExp.dat" using
    1:3 with points pt 2

set output
```

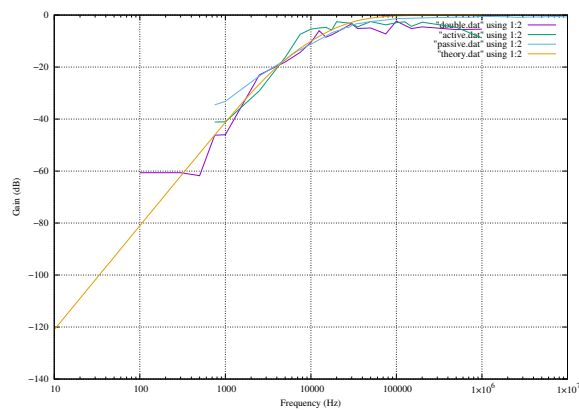


Magnitude response of the double-stage passive filter

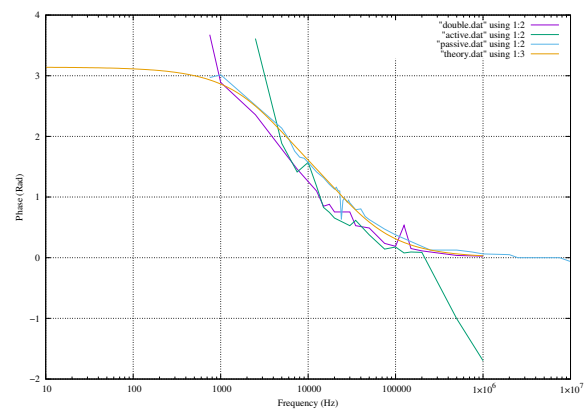


Phase response of the double-stage passive filter

Active v/s Passive Plots



Gain of double-stage(no correction), active, passive and theoretical double stage



Phase of double-stage(no correction), active, passive and theoretical double stage

Task 3

Aim

To design and implement a band-pass series RLC filter with cut-off frequencies determined by the month of birth of lab partner_2. The objectives include:

- Implementing the series RLC band-pass filter with $f_L = \text{Month of birth of lab partner}_2$ and $f_H = 10 \times (\text{Month of birth of lab partner}_2)$.
- Measuring and plotting the frequency response (gain and phase) .
- Determining the experimentally obtained centre frequency and bandwidth.
- Comparing experimental results with theoretical calculations and SPICE simulations.

Equipment and Components used

- Multimeter to check resistance and capacitance values
- Resistor of $440\ \Omega$
- Capacitor of $50\ \text{nF}$
- Inductor of $1\ \text{mH}$
- Digital Storage Oscilloscope (Tektronix 1072B) from Tinkerer's Lab
- Function Generator (Tektronix AFG1022)

Procedure

1. Design a series RLC band-pass filter with lower cut-off frequency $f_L = \text{Month of birth of partner}_2$ in kHz and higher cut-off frequency $f_H = 10 \times (\text{Month of birth of lab partner}_2)$ in kHz . The centre frequency is given by

$$f_0 = \frac{1}{2\pi\sqrt{LC}}, \quad \text{Bandwidth} = \frac{R}{2\pi L}.$$

2. Build the series RLC band-pass circuit on the breadboard and apply a sinusoidal input signal.
3. Vary the input frequency from $10\ \text{Hz}$ to $200\ \text{kHz}$ in suitable steps.
4. At each frequency, measure the input and output amplitudes and phases using the DSO.
5. Compute the gain (in dB) and phase shift for each frequency point. Identify the experimentally obtained lower cut-off frequency (f_L), higher cut-off frequency (f_H), centre frequency (f_0), and bandwidth.
6. Simulate the band-pass filter circuit in ngspice. Compare simulated frequency response (gain and phase) with experimental results.

7. Adjust simulation parameters (R, C values) to obtain best fit with experimental data. (We used **MSE regression** to optimize the fit.)

Stage 1 : Circuit Design and Setup

The circuit implemented is a series RLC band-pass filter with $R = 440 \Omega$, $L = 1 \text{ mH}$, and $C = 50 \text{ nF}$.

The centre (resonant) frequency is

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \\ \approx 22.51 \text{ kHz.}$$

The half-power (3 dB) frequencies are obtained from

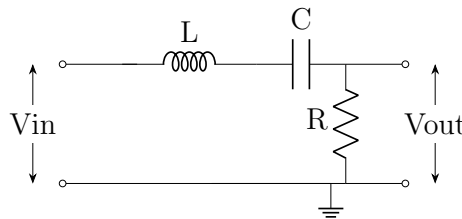
$$(\omega L - \frac{1}{\omega C}) = \pm R,$$

which gives

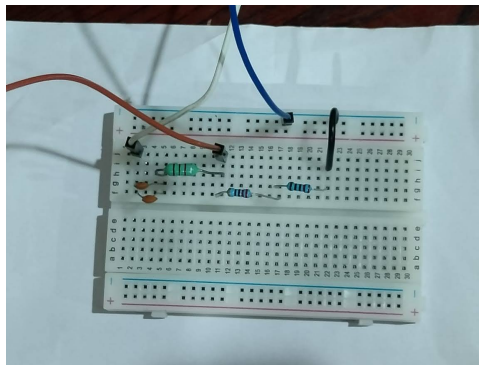
$$\omega_{H,L} = \frac{\sqrt{R^2 + \frac{4L}{C}} \pm R}{2L}, \quad f_{H,L} = \frac{\omega_{H,L}}{2\pi}.$$

Evaluating,

$$f_L \approx 6,610.33 \text{ Hz}, \\ f_H \approx 76,638.51 \text{ Hz}.$$



Breadboard Implementation



Breadboard implementation of the RC high-pass filter.

Observations

Freq	Vout	Vin	Gain	Gain(theo)	Gain error(%)	20log—H—	20log—H'theo—	delT	delPhase	delPhase(Theo)	Phase error(%)
10	0	5	0	0.00138229972	100.00	#NUM!	-57.1879556		0	1.569414027	100
50	0	5	0	0.006911372871	100.00	#NUM!	-43.20871352		0	1.563884899	100
100	0.144	5.2	0.02769230769	0.01382196003	100.35	-31.15281703	-37.18860735	2.4ms	1.507964474	1.556973927	3.147737548
250	0.208	5.12	0.040625	0.03454115924	17.61	-27.82413252	-29.23326182	840us	1.319468915	1.536248295	14.1109599
500	0.376	5.12	0.0734375	0.0689844306	6.46	-22.68164232	-23.22497832	480us	1.507964474	1.501757064	0.4133431184
750	0.512	5.12	0.1	0.103233276	3.13	-20	-19.72360581	320us	1.507964474	1.467378804	2.765861765
1000	0.68	5.04	0.1349206349	0.1371938137	1.66	-17.39843247	-17.25330943	230us	1.445132621	1.433168446	0.8348058815
2000	1.28	4.96	0.2580645161	0.2684330689	3.86	-11.76543414	-11.42327967	100us	1.256637061	1.299030296	3.263452341
3000	2	4.96	0.4032258065	0.3889472867	3.67	-7.889033617	-8.202185076	52us	0.9801769079	1.171307699	16.31772684
4000	2.4	4.96	0.4838709677	0.4958272759	2.41	-6.305408696	-6.09339172	44us	1.105840614	1.052009133	5.117016477
5000	2.6	4.88	0.5327868852	0.588042101	9.40	-5.468929481	-4.611831588	32us	1.005309649	0.942160277	6.702614589
6000	3	4.8	0.625	0.6660027109	6.16	-4.082399653	-3.530480062	24us	0.9047786842	0.8419591063	7.461119841
7000	3.2	4.72	0.6779661017	0.730989429	7.25	-3.375840406	-2.721778068	20us	0.879645943	0.7510255508	17.12596755
7500	3.12	4.72	0.6610169492	0.7591317527	12.92	-3.595748092	-2.393656853	16us	0.7539822369	0.7088180934	6.371753746
8000	3.4	4.72	0.7203389831	0.7846628911	8.20	-2.849261632	-2.106337718	15us	0.7539822369	0.6686441789	12.76285065
9000	3.6	4.72	0.7627118644	0.8287360018	7.97	-2.352789957	-1.631675884	12us	0.6785840132	0.5939510267	14.24915232
10000	3.6	4.64	0.775862069	0.864793929	10.28	-2.204309596	-1.261747357	9us	0.5654866776	0.5260564964	7.495427101
12500	3.8	4.64	0.8189655172	0.9283980235	11.79	-1.734687679	-0.6453158556	6us	0.471238898	0.3807181394	23.77631884
15000	4	4.56	0.8771929825	0.9658922498	9.18	-1.138097027	-0.3014263726	4us	0.3769911184	0.2619290859	43.92869627
17500	3.92	4.56	0.8506491228	0.9868975857	12.89	-1.313575513	-0.1145582694	1.8us	0.1979203372	0.1620563292	22.13058149
20000	4	4.56	0.8771929825	0.9971156704	12.03	-1.138097027	-0.0250891687	1.3us	0.163362818	0.07596995754	115.0360791
22500	4	4.48	0.8928571429	0.9999999745	10.71	-0.9843604534	-0.00000221602201	500ns	0.07068583471	0.002258888929	31.192.30206
25000	4	4.56	0.8771929825	0.997721139	12.08	-1.138097027	-0.01981652329	0	0	-0.06752373008	100
27500	4.08	4.56	0.8947368421	0.9917020729	9.78	-0.9660935915	-0.07237557925	0	0	-0.1289141454	100
30000	4	4.48	0.8928571429	0.9829157528	9.16	-0.9843604534	-0.1496740921	-400ns	-0.07539822369	-0.1851114004	59.26873033
35000	3.98	4.48	0.8883928571	0.9596283751	7.42	-1.027898838	-0.3579383787	-800ns	-0.1759291886	-0.2851183353	38.29608032
40000	3.92	4.56	0.8596491228	0.9315470545	7.72	-1.313575513	-0.6159040606	-1us	-0.2513274123	-0.3721518252	32.46643029
45000	3.84	4.56	0.8421052632	0.900876823	6.52	-1.492672366	-0.9066917221	-1.2us	-0.3392920066	-0.4490110457	24.43571047
50000	3.76	4.56	0.8245614035	0.8690343734	5.12	-1.675539955	-1.219260906	-1.2us	-0.3769911184	-0.5175491042	27.15838645
55000	3.68	4.56	0.8070175439	0.8369623306	3.58	-1.86234048	-1.545881759	-1.4us	-0.4838052687	-0.5790875782	16.45386866
60000	3.6	4.64	0.775862069	0.8052926845	3.65	-2.204309596	-1.880924923	-1.4us	-0.5277875658	-0.6346273562	16.83504333
62500	3.6	4.64	0.775862069	0.7897458442	1.76	-2.204309596	-2.050253016	-1.4us	-0.5497787144	-0.6604017563	16.75087034
65000	3.52	4.56	0.7719298246	0.7744433372	0.32	-2.248443584	-2.220207046	-1.4us	-0.571769863	-0.6849616063	16.52526833
70000	3.44	4.56	0.7543859649	0.744680998	1.30	-2.448128002	-2.560594558	-1.5us	-0.6597344573	-0.7307395758	9.716884218
72500	3.36	4.64	0.724137931	0.7302604676	0.84	-2.803574063	-2.730444183	-1.4us	-0.6377433087	-0.75209319	15.20421709
75000	3.36	4.64	0.724137931	0.7161648012	1.11	-2.803574063	-2.899740559	-1.4us	-0.6597344573	-0.772504712	14.59800219
76000	3.36	4.64	0.724137931	0.7106194521	1.90	-2.803574063	-2.967258171	-1.4us	-0.6685309167	-0.7804180753	14.33682307
77000	3.28	4.64	0.7068965517	0.7051278009	0.25	-3.012882737	-3.034643243	-1.5us	-0.725707903	-0.7881929642	7.927634777
77500	3.28	4.64	0.7068965517	0.7024022059	0.64	-3.012882737	-3.068282664	-1.3us	-0.6330309197	-0.7920294989	20.07483047
78000	3.28	4.64	0.7068965517	0.6996901389	1.03	-3.012882737	-3.101884935	-1.5us	-0.7351326809	-0.7958326303	7.627225501
79000	3.2	4.64	0.6896551724	0.6943066851	0.67	-3.227360045	-3.168973062	-1.4us	-0.694920295	-0.8033402407	13.49614276
80000	3.2	4.64	0.6896551724	0.6889775909	0.10	-3.227360045	-3.235898067	-1.4us	-0.7037167544	-0.8107188639	13.19842355
85000	3.12	4.72	0.6610169492	0.6631486921	0.32	-3.595748092	-3.567781652	-1.4us	-0.7476990516	-0.8457786254	11.59636468
90000	3.04	4.72	0.6440677966	0.6386705428	0.85	-3.821368301	-3.894462284	-1.4us	-0.7916813487	-0.8780270365	9.834057972
95000	3.04	4.72	0.6440677966	0.6155108119	4.64	-3.821368301	-4.215286279	-1.3us	-0.7759733854	-0.9077623913	14.51800682
100000	2.96	4.72	0.6271186441	0.5936211711	5.64	-4.053005752	-4.529812373	-1.3us	-0.8168140899	-0.9352451375	12.6631022
150000	2.32	4.8	0.4833333333	0.4309761553	12.15	-6.31506505	-7.31093515	-1.12us	-1.055575132	-1.125222053	6.189615718
200000	1.84	4.96	0.3709677419	0.3342370944	10.99	-8.61327707	-9.518907055	-880ns	-1.105840614	-1.230000671	10.09430809
250000	1.52	4.96	0.3064516129	0.2717725844	12.76	-10.27276177	-11.31588711	-740ns	-1.162389282	-1.295561862	10.27913711
500000	0.8	4.96	0.1612903226	0.1389787832	16.05	-15.84783379	-17.1410299	-420ns	-1.319468915	-1.431366211	7.817516996
750000	0.48	4.88	0.09836065574	0.09304961135	5.71	-20.14357169	-20.62570873	-310ns	-1.460840584	-1.477611915	1.135029524
1000000	0.344	4.88	0.07049180328	0.06989233179	0.86	-23.03722759	-23.1114094	-210ns	-1.319468915	-1.500846966	12.08504637
2000000	0.192	4.88	0.0393442623	0.0349970708	12.42	-28.10237187	-29.11936608	-160ns	-2.010619298	-1.535792108	30.91741309

Experimental lower-cutoff frequency, upper-cutoff frequency and center frequency

Center Frequency: The center frequency is defined as the frequency at which the gain is maximum.

From the data:

$$f_c = 27.5 \text{ kHz}$$

Lower and Upper Cutoff Frequencies: The cutoff frequencies are defined as the points where the gain drops by 3 dB from the maximum value. From the experimental gain vs. frequency data, two such points are identified as the lower cutoff frequency (f_L) and the upper cutoff frequency (f_H).

$$f_L = 6 \text{ kHz}, \quad f_H = 90 \text{ kHz}$$

Band-width The band-width can be calculated from the cutoff frequencies using the below relation:

$$BW = f_h - f_L$$

Substituting the experimental values of f_L and f_H :

$$BW = 84 \text{ kHz}$$

Center Frequency from Cutoff Points: The center frequency can also be calculated from the cutoff frequencies using the below relation:

$$f_c = \sqrt{f_L \cdot f_H}$$

Substituting the experimental values of f_L and f_H :

$$f_c = \sqrt{6 \times 90} \approx 23.2 \text{ kHz}$$

Parameter	Experimental Value	Theoretical Value
Lower Cut-off Frequency, f_L	6 kHz	6.6 kHz
Upper Cut-off Frequency, f_H	90 kHz	76.6 kHz
Center Frequency, f_c	$\approx 23.2 \text{ kHz}$	22.51 kHz
Bandwidth, BW	$\approx 84 \text{ kHz}$	70 kHz

Note

The center frequency obtained directly from the maximum gain (27.5 kHz) differs from the value computed using the cutoff frequencies (23.2 kHz). This deviation might be due to measurement uncertainties

Simulation using SPICE

We have simulated the filter using `ngspice` and plotted the output using `gnuplot`.

```
*Netlist
Vin in 0 AC 5
L in x 1m
C x out 50nF
R out 0 440

.control
AC DEC 30 10 200k
set wr_singlescale
wrdata band.dat 20*log10(abs(v(out)/v(in))) phase(v(out))
.endc
.end
```

```

set terminal postscript eps enhanced color font 'Times,12'
set logscale x
set output 'bpassexp.eps'

set xlabel 'Frequency (Hz)'
set ylabel 'Gain (dB)'
set grid

plot "band.dat" using 1:2 with lines, "bandexp.dat" using 1:2
    with points pt 2

set output

```

```

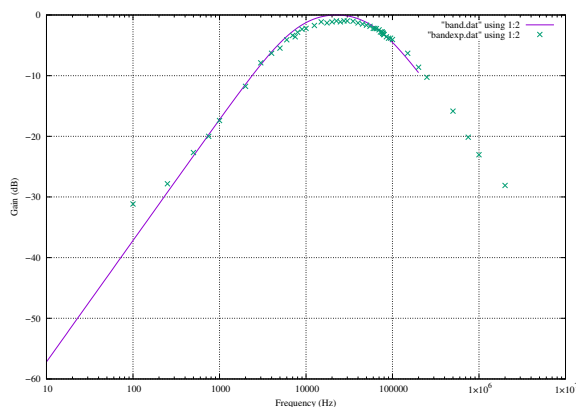
set terminal postscript eps enhanced color font 'Times,12'
set logscale x
set output 'bandphexp.eps'

set xlabel 'Frequency (Hz)'
set ylabel 'Phase (Rad)'
set grid

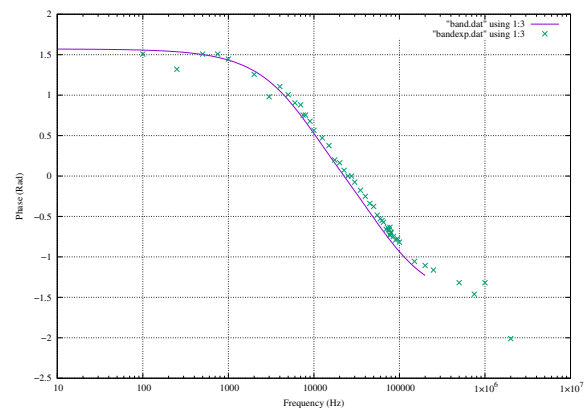
plot "band.dat" using 1:3 with lines, "bandexp.dat" using 1:3
    with points pt 2

set output

```



Magnitude response of the RLC band-pass filter



Phase response of the RLC band-pass filter

Closest fit for experimental data

The experimental gain and phase responses for RLC band-pass filters were compared against simulation models. Mean squared error (MSE) minimization technique was used to find the optimal RC value that minimized the mean squared error.

```

import pandas as pd
import numpy as np

```

```

import matplotlib.pyplot as plt
pi = np.pi

# Load Bandpass filter data from CSV (skipping first 2 and last
  row for cleanup)
df = pd.read_csv('Task1_EE2501 - Bandpass.csv')
df = df.iloc[2:-1]

# Extract frequency, gain (experimental/theoretical), and phase (
  experimental/theoretical)
freqs = np.array(df['Freq(num)'])
H_exp = np.array(df['Gain'])
H_theo = np.array(df['Gain(theo)'])
phase_exp = np.array(df['delPhase'])
phase_theo = np.array(df['delPhase(Theo)'])

# Compute Mean Squared Error (MSE) between experimental and
  theoretical values
def calculateMSE(expe, theo):
    if (len(expe) != len(theo)):
        raise ValueError("Experimental and theoretical data
          lengths do not match.")
    return np.mean([(expe[i] - theo[i])**2 for i in range(len(
      expe))])

# Transfer function magnitude  $|H(j \omega)|$  of an RLC bandpass filter
  for given RC and LC
def H_bandpass(fs, RC, LC):
    omegas = 2 * pi * fs
    num = omegas * RC
    den = np.sqrt((omegas * RC)**2 + (1 - LC * omegas**2)**2)
    return num / den

# Phase response  $\angle H(j \omega)$  of an RLC bandpass filter for given RC
  and LC
def phase_bandpass(fs, RC, LC):
    omegas = 2 * pi * fs
    num = 1 - omegas**2 * LC
    den = omegas * RC
    return np.arctan(num / den)    # consider np.arctan2(num, den)
    for correct quadrant

# Circuit parameters: R = 440  $\Omega$ , C = 50 nF, L = 1 mH
RC_init = 440 * 50e-9    # initial RC product
LC = 1e-3 * 50e-9        # LC product

# Search parameters for fitting RC by minimizing MSE
step = 1e-3
RCs = []
MSEs = []

```

```

# Sweep RC scaling factor from 'step' to 100 RC_init , compute
MSE for each
for i in np.arange(step, 100, step):
    RC = RC_init * i
    H_RC_theo = H_bandpass(freqs, RC, LC)
    RCs.append(RC)
    MSEs.append(calculateMSE(H_exp, H_RC_theo))

# Find RC corresponding to the minimum MSE
min_MSE_index = np.argmin(MSEs)
print(f"RC corresponding to minimum MSE is {RCs[min_MSE_index]}
      which is {step * (min_MSE_index + 1)} times of initial RC.")

# Print MSE comparison for Gain (initial theoretical vs optimized
RC fit)
print(f"For Gain => Initial MSE: {calculateMSE(H_exp, H_theo)},
      Minimum MSE: {calculateMSE(H_exp, H_bandpass(freqs, RCs[
min_MSE_index], LC))}")

# Plotting 20logH with experimental, theoritical with initial RC
and RC with minimum MSE
plt.figure(figsize=(14,6))
plt.semilogx(freqs, 20*np.log10(H_exp), label="Experimental")
plt.semilogx(freqs, 20*np.log10(H_theo), label="Theoretical")
plt.semilogx(freqs, 20*np.log10(H_bandpass(freqs, RCs[
min_MSE_index], LC)), label="Theoretical with min. MSE")
plt.xlabel("Frequencies")
plt.ylabel("20log|H|")
plt.legend()
plt.grid()
plt.show()

# Print MSE comparison for Phase (initial theoretical vs
optimized RC fit)
print(f"For Phase => Initial MSE: {calculateMSE(phase_exp,
phase_theo)}, Minimum MSE: {calculateMSE(phase_exp,
phase_bandpass(freqs, RCs[min_MSE_index], LC))}")

# Plotting Phase with experimental, theoritical with initial RC
and RC with minimum MSE
plt.figure(figsize=(14,6))
plt.semilogx(freqs, phase_exp, label="Experimental")
plt.semilogx(freqs, phase_theo, label="Theoretical")
plt.semilogx(freqs, phase_bandpass(freqs, RCs[min_MSE_index], LC)
, label="Theoretical")
plt.xlabel("Frequencies")
plt.ylabel("Phase")
plt.legend()
plt.grid()
plt.show()

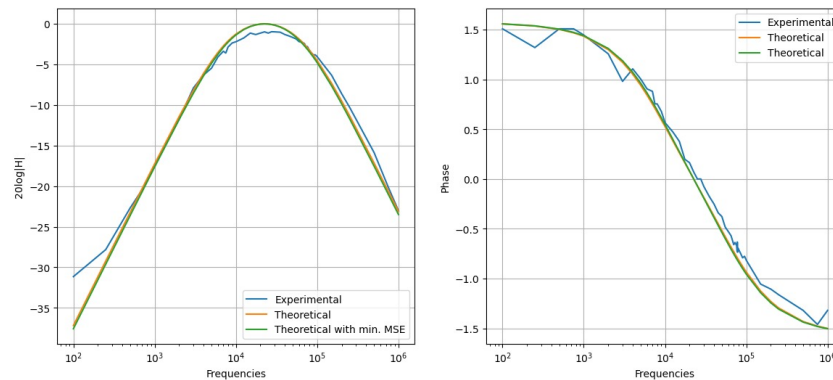
```

By running the above program, we obtained the best-fit values of RC and the corresponding cutoff frequencies f_c for the bandpass RLC filter:

- **Bandpass RLC Filter:**

$$RC_{\min \text{ MSE}} = 2.1032 \times 10^{-5} \text{ s} \quad (0.956 \times \text{initial } RC)$$

The frequency response (gain) and phase plots for the best-fit, experimental, and theoretical data are shown below:



Bandpass RLC Filter: Experimental, Theoretical, and Best-fit frequency response and phase plots.

Note :

Considering space limitations in the report, we uploaded the pictures of readings of DSO [here](#).

Plagiarism Check

Turnitin software was used for plagiarism check for the report, snapshot of originality report can be found below :

